Signal Lost (Remote Facility / Communications Degradation & Electronic Warfare)

Executive Summary

Signal Lost is a **remote communications-facility battlefield** built for **electronic warfare, comms degradation, autonomous counter-EW behavior, and precision operations across isolated terrain**. The environment features satellite dishes, microwave towers, buried fiber lines, hardened relay bunkers, diesel generators, snow-scoured ridges or dry scrub flats (depending on season), and long-range sensor arrays. The map emphasizes Obsidian Protocol's autonomous EW-aware routing, sensor-fusion under jamming and interference, comms-relay warfare, and logistics across a remote, infrastructure-dependent battlespace. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of remote-facility terrain. Key features include EW zones, blackout corridors, relay-tower control, and comms-dependent maneuver warfare.

Design Goals & Gameplay

Communications Degradation Warfare

Signal Lost prioritizes jamming, spoofing, interference, blackout zones, and comms-dependent tactics. Units operate under partial or total signal loss.

Electronic Warfare (EW) Dominance

Players deploy EW vehicles, drone jammers, signal scramblers, and counter-EW units. Destroying or capturing relay towers shifts the entire tactical landscape.

Autonomous EW-Aware Routing

Autonomous units choose routes based on signal strength, interference pockets, terrain occlusion, and fallback relay availability.

Remote-Facility Combat

Long-range fire, drone reconnaissance, armored pushes, and stealth infiltration revolve around controlling the facility's communication backbone.

Logistics & Resources

Fuel caches, generator stations, and buried cable hubs serve as resupply nodes. Convoys must navigate blackout corridors and avoid EW-driven ambush zones.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between EW dominance, stealth infiltration, or signal-resistant armor. Blue deploys from the southwest service-road checkpoint; Red deploys from the northeast relay-spire command.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² remote-facility zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Service-Road Checkpoint) — (0,0), 150x150 m Armored vehicles, EW trucks, drone pads.

Red HQ (Relay-Spire Command) — (1000,1000), 150x150 m Elevated bunker with radar mast, EW arrays, AA emplacements.

Central Relay Basin — (500,450), 300x300 m Relay towers, interference pockets, blackout zones, long-range EW duels.

Generator Station Cluster — (250,800), 120x120 m Diesel generators, transformers, interior flanking routes.

Sensor Array Ridge — (800,300), 200x200 m High-ground overwatch, signal triangulation, counter-EW vantage points.

Blackout Corridor — (0,450 → 400,900) Primary EW chokepoint; jamming fields, interference pockets, false signals.

Service Road Route — (300,0 → 450,350) Safer but predictable; vulnerable to EW-driven ambushes.

Relay Tower A — (650,150), +240 m Long-range sensor vantage point.

Relay Tower B — (150,650), +260 m Comms relay hub and EW-control node.

Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Blackout Corridor

Major chokepoint; filled with jamming fields and false signals.

Sensor Array Ridge

Extreme long-range sightlines; ideal for counter-EW and recon triangulation.

Relay Basin

Long-range EW engagements distorted by interference and signal loss.

Service Road Route

Predictable but safe; vulnerable to EW-driven ambushes.

Sightlines

Relay Tower A provides long-range sensor coverage (~350 m). Interference and jamming create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW service-road checkpoint + EW vehicles.

Red Deployment: NE relay-spire command + AA emplacements.

Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Ridges, flats, relay towers, generator stations, service roads.

Vegetation

Sparse shrubs, dry grass, or snow patches depending on climate.

Buildings & Props

Military: EW trucks, AA guns, radar dishes, bunkers. Industrial: Generators, transformers, pipelines, relay towers. Logistics: Trucks, crates, fuel drums, drones.

Effects

Signal distortion, interference waves, smoke, dust, fog.

Audio

Wind, generator hum, static bursts, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Facility/Ri	3	200k	600k
Industrial/	4	80k	320k
Buildings/	15	50k	750k
Roads/Pat	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out relay basin, ridges, generator stations, AR anchor test.

2. Core Systems — Month 2 NavMesh, EW autonomy behaviors, interference logic.

3. AR Integration — Month 3 Anchors, plane detection, basic interactivity.

4. Content Fill — Month 4–5 Final models, textures, lighting, audio.

5. Gameplay & Polish — Month 6–7 Objectives, balance, VFX (interference waves, static bursts).

6. Testing & QA — Month 8 Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify EW-aware navigation, interference-zone avoidance, fallback routing.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under jamming.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, EW logic, deployment budget enforcement, anchor drift, sensor accuracy.